Review Sheet 1: Process Analysis and Improvement[1]

1. Instant Dolls Co.*

Instant Dolls specializes in manufacturing dolls in a make to order basis. Their business process consists on the following activities, which must be conducted in the sequence described below:

Activity 1: Get the specifications of the doll from the customer, get the payment and submit the order to the production line. (8 minutes)

Activity 2: Assemble the main body of the doll (10 minutes)

Activity 3: Add hair and paint. (12 minutes)

Activity 4: Dress the doll (7 minutes)

Activity 5: Package the doll (3 minutes)

(The time required for each activity is shown in parenthesis)

The whole process is conducted by three workers. The assignment of tasks to workers is the following:

W1 does Activity 1.

W2 does Activities 2 and 3.

W3 does Activities 4 and 5.

a) What is the bottleneck of the process?

b) How much time will it take to produce 250 units?

c) What is the utilization of worker 3, when the plant is operating at full capacity?

d) What is the average labor utilization of the workers? Assume the process operates at its capacity and there are no empty system effects.

[1] Problems are rated for difficulty: * = Relatively easy, for first review of concepts; ** = Challenging, assumes good understanding of the concepts; *** = Advanced, assumes full understanding of the concepts and creativity in application.

e) What are the direct labor costs for one doll? Assume a wage rate of $15 per hour.

To increase the production rate, Instant Dolls is considering two alternatives:

Alternative I: To hire a new worker to help any of the workers without changing the tasks performed by each worker.

Alternative II: To redesign the assignment of tasks to workers. For this, the company is evaluating to re-assign Step 2 from W2 to W1.

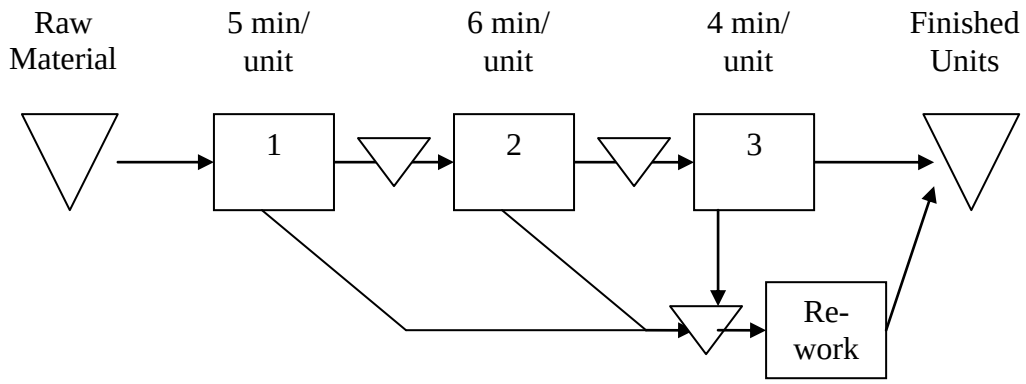
f) Suppose that Alternative I is chosen. Where would you assign the new employee?

g) What would be the direct labor of one doll under the solution depicted above? Assume there is sufficient demand (system operates at its capacity) and there are no empty system effects.

h) What would be the direct labor costs for Alternative II?

2. Ceramics Production**

Will Clintoon manages a glass ceramics production process that has three stages. Each stage operates at the rate that the operator at each stage can work. Squares represent process steps, and triangles represent the potential for work in process inventory.



The process is experiencing serious quality problems related to management providing insufficient training for workers. Approximately 20% of the parts going through stage 1 are badly processed by the operator. Instead of scrapping the unit, it is sent to a highly skilled rework operator that can correct the mistake and finish the unit completely in 15 minutes.

The same problem occurs at stage 2 where 10% of the parts are poorly processed, and require 10 minutes of rework. Stage 3 also has a 10% defect rate, each defect requires 5 minutes of the rework operators time.

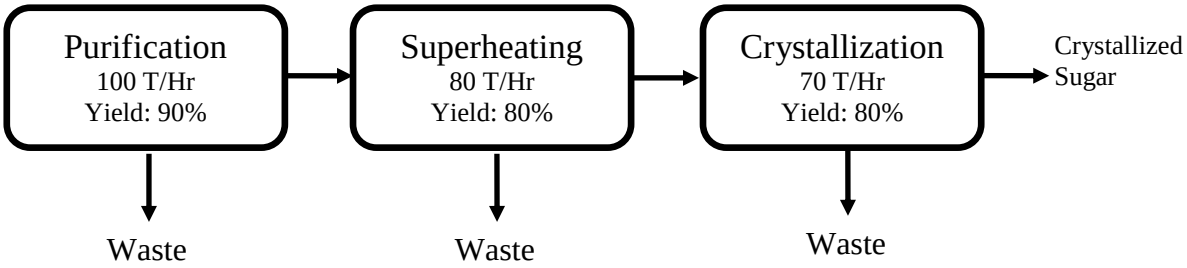
(i) What is the utilization of stage 2 if work is released into the system at a rate of 5 units per hour?

(ii) Where in the process is the bottleneck (i.e., which resource is most highly utilized under any sustainable work release rate, that is a rate that does not lead to continuous buildupof inventory)? Why?

(iii) After a training program is implemented, the defect rate at stage 1 is reduced to 10% and the defect rates at stage 2 and stage 3 are both reduced to 5%. What is the maximum process capacity after these changes?

3. A Sweet Job **

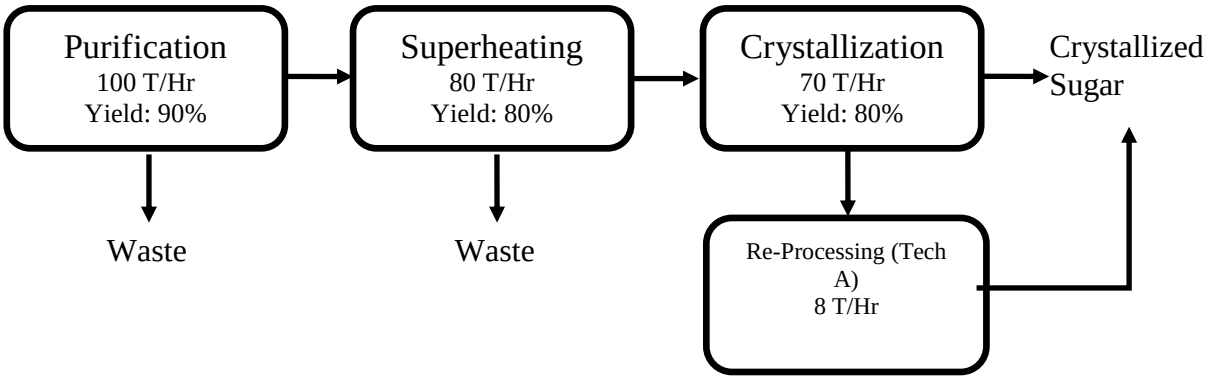
In an unexpected twist of events, you recently inherited a sugar plant from a distant relative. You are looking forward to apply your extensive knowledge of POM principles to this plant.
On your first visit to the sugar producing facility, you learn the following. Sugar production has basically three steps- Purification of Molasses followed by Superheating of the purified molasses and finally crystallization. Your facility has three extremely reliable continuous processing machines for the three process steps. To ensure the freshness of sugar, at no point in the process, can one build work in process inventories. The performance of the individual processing machines is described below.
Step 1, Purification: Capacity- 100 Tons/hour (T/hr) Yield- 90% (in other words, if 100 Tons of input is introduced in the machine, only 90 tons is available for the next step)
Step 2, Superheating: Capacity- 80 Tons/hour. Yield -80%
Step 3, Crystallization: Capacity- 70 Tons/hour. Yield- 80%
The process is illustrated below:



a) ***If 50 T/hr of molasses are introduced into the purification stage, what would be the utilization of each of the stages? What stage is the bottleneck? What is the capacity of this plant (in Tons of crystallized sugar per hour)? ****

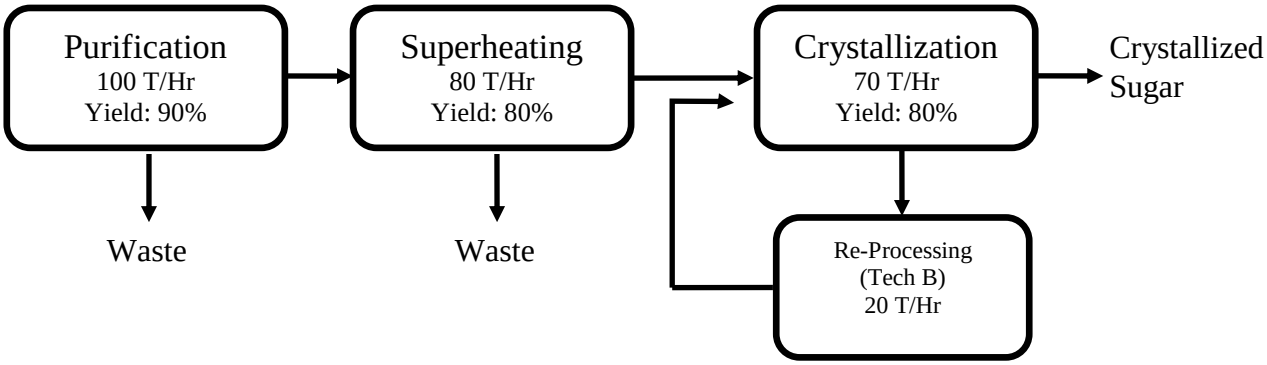
After analyzing the plant, you feel a little frustrated by the high yield losses. You contact your classmate from INSEAD, who is now the leading process improvement consultant for the sugar industry (What a sweet job!). He points out that there are a couple of new technologies available that can re-process the losses from the crystallization stage. This could increase the total capacity of the plant.

Technology A can take the losses from the crystallization stage and convert all the waste to crystallized sugar (the final product). (For efficient operation of the re-processing stage, it must operate on all the waste from the crystallization stage. In other words, ***all*** the waste that comes from the crystallization stage must be introduced into the Re-processing stage.) The modified Process diagram is illustrated below.



b) If you buy a Technology-A Re-processing machine with a capacity of 8 T/Hr, what is the new capacity of the plant? What is the minimum capacity of the new machine that would not change the bottleneck from the one you found in Part A above? **

Technology B on the other hand takes the losses from the crystallization stage and converts them to super-heated liquid, which must be again re-introduced into the crystallization stage. The process diagram is illustrated below



c) ***If you buy a Technology-B Re-processing machine with a capacity of 20 T/Hr, what is the new capacity of the plant? (Like Technology A, efficient operation of Technology B requires that all the waste from the crystallization stage be introduced into the re-processing stage) ******

4. Sport Obermeyer *

Sport Obermeyer manufactures parkas using an assembly line. It owns two factories, one in Mexico and the other in China. The workers in the Chinese factory are multi-skilled. The workers in Mexico are relatively less skilled and more specialized. Sport Obermeyer uses an assembly line with twenty workers in China and one with 40 workers in Mexico. The capacities are the same in the two assembly lines. Based on this information you can conclude that relative to the factory in China (explain to get credit):

a. The Mexican factory makes more parkas per hour but takes less time to make parkas.
b. The Mexican factory makes more parkas per hour but takes longer to make parkas.
c. The Mexican factory makes the same number of parkas per hour but takes longer to make parkas.
d. The Mexican factory makes the same number of parkas per hour but takes the same time to make parkas.
e. None of the above

(Note: Time to make a parka means the time from beginning to end of the assembly line. Assume that the transfer lot size is one, that is when a worker finishes one operation the garment is passed to the next worker for the next operation.)